

Date of Birth
01/11/1966

Sex
Female

Physician
Krishna Alluri

Institution
Texas Oncology San Antonio
Northeast
11200598

TEMPUS | xF+
523 gene liquid biopsy

cfDNA specimen:
Peripheral Blood
Collected 10/18/2024
Received 10/22/2024

GENOMIC VARIANTS

Potentially Actionable		Variant Allele Fraction
BRCA2	p.F2560fs Frameshift - LOF	48.8%
EGFR	p.E746_S752delinsV Inframe deletion (exon 19) - GOF	1.4%

ctDNA Tumor Fraction

0.3%

ctDNA tumor fraction is a quantitative measure of circulating tumor DNA.

IMMUNOTHERAPY MARKERS

Blood Tumor Mutational Burden (bTMB)	Microsatellite Instability Status
5.0 m/MB	MSI-High not detected

A bTMB of 15 m/MB most closely corresponds to a TMB of 10 m/MB measured with the Tempus xT tissue assay. Tissue-based testing is recommended to determine the appropriateness of immunotherapy.

FDA-APPROVED THERAPIES, CURRENT DIAGNOSIS

EGFR Inhibitor	Afatinib	NCCN, Consensus, Non-Small Cell Lung Cancer MSK OncoKB, Level 1 EGFR p.E746_S752delinsV Exon 19
	Dacomitinib	NCCN, Consensus, Non-Small Cell Lung Cancer MSK OncoKB, Level 1 EGFR p.E746_S752delinsV Exon 19
	Erlotinib	NCCN, Consensus, Non-Small Cell Lung Cancer MSK OncoKB, Level 1 EGFR p.E746_S752delinsV Exon 19
	Gefitinib	NCCN, Consensus, Non-Small Cell Lung Cancer MSK OncoKB, Level 1 EGFR p.E746_S752delinsV Exon 19

EGFR Inhibitor	Osimertinib	 NCCN, Consensus, Non-Small Cell Lung Cancer  MSK OncoKB, Level 1 EGFR p.E746_S752delinsV Exon 19
Anti-EGFR/MET BsAb	Amivantamab-vmjw	 NCCN, Consensus, Non-Small Cell Lung Cancer  MSK OncoKB, Level 1 EGFR p.E746_S752delinsV Exon 19
Combination (EGFR Inhibitor + VEGFR2 Inhibitor)	Erlotinib + Ramucirumab	 NCCN, Consensus, Non-Small Cell Lung Cancer  MSK OncoKB, Level 1 EGFR p.E746_S752delinsV Exon 19
Combination (EGFR Inhibitor + Anti-EGFR/MET BsAb)	Lazertinib + Amivantamab-vmjw	 NCCN, Consensus, Non-Small Cell Lung Cancer  MSK OncoKB, Level 1 EGFR p.E746_S752delinsV Exon 19

✓ FDA-APPROVED THERAPIES, OTHER INDICATIONS

Combination (EGFR Inhibitor + Anti-EGFR MAb)	Afatinib + Cetuximab	 NCCN, Consensus, Non-Small Cell Lung Cancer EGFR p.E746_S752delinsV Exon 19
PARP Inhibitor	Niraparib	NCCN, Consensus, Ovarian Cancer BRCA2 p.F2560fs Loss-of-function
	Olaparib	NCCN, Consensus, Breast Cancer BRCA2 p.F2560fs Loss-of-function
	Rucaparib	NCCN, Consensus, Prostate Cancer BRCA2 p.F2560fs Loss-of-function
	Talazoparib	NCCN, Consensus, Breast Cancer BRCA2 p.F2560fs Loss-of-function
Combination (PARP Inhibitor + Anti-VEGF MAb)	Niraparib + Bevacizumab	NCCN, Consensus, Ovarian Cancer BRCA2 p.F2560fs Loss-of-function
	Olaparib + Bevacizumab	NCCN, Consensus, Ovarian Cancer BRCA2 p.F2560fs Loss-of-function

We were unable to determine whether treatments on this report were previously prescribed for this patient.

Study to Assess Adverse Events and Change in Disease Activity in Adult Participants With Advanced Solid Tumors Receiving Intravenous (IV) ABBV-400 as Monotherapy and in Combination With IV Bevacizumab (NCT05029882)	Phase I San Antonio, TX - 8 mi ✓ EGFR p.E746_S752delinsV mutation
A Study to Investigate the Safety and Efficacy of KQB198 as Monotherapy and in Combination in Participants With Advanced Solid Malignancies (NCT06507306)	Phase I San Antonio, TX - 8 mi ✓ EGFR p.E746_S752delinsV mutation
A Study of PARG Inhibitor ETX-19477 in Patients With Advanced Solid Malignancies (NCT06395519)	Phase I Houston, TX - 182 mi ✓ BRCA2 p.F2560fs mutation

VARIANTS OF UNKNOWN SIGNIFICANCE

Gene	Mutation effect	Variant allele fraction
NOTCH2	c.7_8delinsTT p.A3F Missense variant NM_024408	15.7%
KMT2C (MLL3)	c.2872G>C p.D958H Splice region variant NM_170606	6.7%
KMT2C (MLL3)	c.938G>A p.G313E Missense variant NM_170606	4.2%
SLFN11	c.2044G>A p.G682R Missense variant NM_152270	3.0%

LOW COVERAGE REGIONS

CARM1	CUL4A	CYP17A1	FANCC	FANCI
IMPDH1	KDM5D	MAPK1	MAPK3	MAX
NFE2L2	NOTCH2	NTRK2	PHLPP2	PIK3R2
PPP6C	PTPRT	RAD51C	RHOA	RXRA
SDHAF2	TERT	TGFBR1	TP63	ZNRF3

VARIANT DETAILS - POTENTIALLY ACTIONABLE

⊕

BRCA2

c.7679_7680del p.F2560fs NM_000059 Frameshift - LOF

VAF: 48.8%

BRCA2 encodes a nuclear phosphoprotein which helps maintain DNA stability through homologous recombination based DNA double stranded break repair and involvement in DNA damage checkpoint control. Loss of function mutations and copy number loss of BRCA2 are associated with cancer progression.

Assay Description (continued)

1. Nivolumab plus Ipilimumab in Lung Cancer with a High Tumor Mutational Burden. <https://www.ncbi.nlm.nih.gov/pubmed/29658845>
2. Cancer immunology. Mutational landscape determines sensitivity to PD-1 blockade in non-small cell lung cancer. <https://www.ncbi.nlm.nih.gov/pubmed/25765070>

ctDNA tumor fraction is a quantitative measure of circulating tumor DNA in a cancer patient's blood sample determined from analysis of CNV and VAF values, excluding common CH variants. The ctDNA tumor fraction value for a patient may differ from a value estimated from VAF or CNV data only. Variants and CNVs used in this calculation may not necessarily appear on this report. The ctDNA tumor fraction feature is not included in reports that are CE-IVD marked under IVDD.

xF+ provides insights into the clinically relevant biomarkers incorporated in OncoKB, NCCN and other oncology guidelines for:

Bladder cancer: FGFR2, FGFR3

Breast cancer: BRCA1, BRCA2, ERBB2 (HER2), ESR1, PIK3CA

Cholangiocarcinoma: FGFR2, IDH1

Colorectal cancer: BRAF, ERBB2 (HER2), KRAS, NRAS

Gastroesophageal adenocarcinoma: ERBB2 (HER2)

Gastrointestinal stromal tumor: KIT, PDGFRA

Melanoma: BRAF, KIT, NRAS

Non-small cell lung cancer: ALK, BRAF, EGFR, ERBB2 (HER2), KRAS, MET, RET, ROS1

Prostate cancer: ATM, BARD1, BRCA1, BRCA2, BRIP1, CDK12, CHEK1, CHEK2, FANCL, PALB2, RAD51B, RAD51C, RAD51D, RAD54L

Tumor-agnostic: NTRK1, NTRK2, NTRK3

Complete Gene List**Gene Rearrangements**

ALK, BRAF, FGFR1, FGFR2, FGFR3, NTRK1, NTRK2, NTRK3, RET, ROS1

Copy Number Gains

CCNE1, CD274 (PD-L1), EGFR, ERBB2 (HER2), MDM2, MET, MYC

Genes with Enhanced Coverage

A-C

ABL1, AKT1, AKT2, ALK, APC, AR, ARAF, ARID1A, ATM, ATR, B2M, BAP1, BRAF, BRCA1, BRCA2, BTK, CCND1, CCND2, CCND3, CCNE1, CD274 (PD-L1), CDH1, CDK12, CDK4, CDK6, CDKN2A, CHEK2, CRKL, CTNNB1

D-F

DDR2, EGFR, ERBB2 (HER2), ERBB3, ERFF1, ESR1, EZH2, FBXW7, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, FOXL2

G-M

GATA3, GNA11, GNAQ, GNAS, HNF1A, HRAS, IDH1, IDH2, JAK1, JAK2, JAK3, KDR, KEAP1, KIT, KMT2A, KRAS, MAP2K1, MAP2K2, MAPK1, MAPK3, MDM2, MET, MLH1, MPL, MSH2, MSH3, MSH6, MTOR, MYC, MYCN, MYD88

N-R

NF1, NF2, NFE2L2, NOTCH1, NPM1, NRAS, NTRK1, NTRK2, NTRK3, PALB2, PBRM1, PDCD1LG2, PDGFRA, PDGFRB, PIK3CA, PIK3R1, PMS2, PTCH1, PTEN, PTPN11, RAD51C, RAF1, RB1, RET, RHEB, RHOA, RIT1, RNF43, ROS1

S-Z

SDHA, SMAD4, SMO, SPOP, STK11, TERT, TP53, TSC1, TSC2, VEGFA, VHL

Assay Description (continued)**Additional Genes****A-C**

ABCC3, ABL2, ABRAXAS1, ACVR1 (ALK2), ACVR1B, AJUBA, AKT3, ALOX12B, AMER1, APLNR, ARFRP1, ARID1B, ARID2, ASNS, ASXL1, ATRX, AURKA, AURKB, AURKC, AXIN1, AXIN2, AXL, BARD1, BAX, BCL2, BCL2L1, BCL2L11, BCL2L2, BCL6, BCLAF1, BCOR, BCORL1, BCR, BIRC3, BLM, BMPR1A, BRD4, BRIP1, BTG1, BTG2, CALR, CARD11, CARM1, CASP8, CBFB, CBL, CD22, CD70, CD74, CD79A, CD79B, CDC73, CDK8, CDK9, CDKN1A, CDKN1B, CDKN2B, CDKN2C, CEBPA, CHD4, CHEK1, CIC, CKS1B, CREBBP, CSF1R, CSF3R, CTC1, CTCF, CTLA4, CTNNA1, CUL3, CUL4A, CUX1, CXCR4, CYLD, CYP17A1, CYSLTR2

D-F

DAXX, DDB2, DDR1, DDX3X, DDX41, DEPTOR, DICER1, DIS3, DNMT1, DNMT3A, DOT1L, DPYD, EBF1, EED, EEF2, EGLN1, EIF1AX, ELF3, EMSY, EP300, EPCAM, EPHA2, EPHA3, EPHB1, EPHB4, ERBB4, ERCC2, ERCC3, ERCC4, ERCC6, ERG, ETNK1, ETV1, ETV4, ETV5, ETV6, EWSR1, EZR, FAM46C, FANCA, FANCC, FANCD2, FANCE, FANCG, FANCI, FANCL, FANCM, FAS, FAT1, FCGR2A, FCGR3A, FGF10, FGF12, FGF14, FGF19, FGF23, FGF3, FGF4, FGF6, FH, FHIT, FLCN, FLT1, FLT4, FOLH1, FOXA1, FOXO1, FOXO3, FOXP1, FRS2, FUBP1

G-M

GABRA6, GALNT12, GATA1, GATA4, GATA6, GID4, GLI2, GNA13, GPC3, GPS2, GREM1, GRIN2A, GRM3, GSK3B, GSTP1, H3F3A, HAVCR2, HDAC1, HDAC2, HGF, HIF1A, HIST1H3B, HLA-B, HNF1B, HOXB13, HSD3B1, HSP90AA1, HSPH1, ID3, IFNA21, IFNAR1, IFNAR2, IFNG, IFNGR1, IFNGR2, IFNW1, IGF1, IGF1R, IKBKE, IKZF1, IL10RA, IL32, IL6R, IL7R, IMPDH1, ING1, INPP4B, INSR, IRF1, IRF2, IRF4, IRS2, JUN, KAT6A, KDM5A, KDM5C, KDM5D, KDM6A, KEL, KLF4, KLHL6, KLLN, KMT2C, KMT2D, LATS1, LCK, LMO1, LRP1B, LTK, LYN, LZTR1, MAF, MALT1, MAP2K4, MAP3K1, MAP3K13, MAP3K21, MAP3K7, MAX, MC1R, MCL1, MDM4, MED12, MEF2B, MEN1, MERTK, MITF, MKNK1, MLH3, MRE11, MS4A1, MST1R, MTAP, MTHFR, MUC16, MUTYH, MYB, MYCL

N-R

NBN, NCOA2, NCOR1, NFKBIA, NKX2-1, NOTCH2, NOTCH3, NOTCH4, NQO1, NRG1, NSD1, NSD2, NSD3, NT5C2, NUTM1, P2RY8, PAK1, PALLD, PARP1, PARP2, PARP3, PAX5, PDCD1, PDK1, PHGDH, PHLPP1, PHLPP2, PIAS4, PIK3C2B, PIK3C2G, PIK3CB, PIK3CD, PIK3CG, PIK3R2, PIM1, PLCG1, PLCG2, PMS1, POLA1, POLD1, POLE, POLQ, POT1, PPARG, PPM1D, PPP2R1A, PPP2R2A, PPP6C, PRDM1, PREX2, PRKACA, PRKAR1A, PRKCI, PRKN, PTK2, PTPN13, PTPRD, PTPRO, PTPRT, QKI, RAC1, RAD21, RAD50, RAD51, RAD51B, RAD51D, RAD52, RAD54L, RARA, RASA1, RBM10, RECQL4, REL, RICTOR, RPS6KB1, RPTOR, RRM1, RSF1, RSPO2, RUNX1, RXRA

S-Z

SDC4, SDHAF2, SDHB, SDHC, SDHD, SETBP1, SETD2, SF3B1, SGK1, SIRPA, SLC34A2, SLC9A3R1, SLFN11, SLIT2, SMAD2, SMAD3, SMARCA2, SMARCA4, SMARCB1, SMC1A, SMC3, SNCAIP, SOCS1, SOS1, SOX2, SOX9, SPEN, SRC, SRSF2, STAG2, STAT3, STAT5B, STAT6, SUFU, SUZ12, SYK, TBX3, TCF7L2, TEK, TERC, TET2, TFEB, TGFB1, TGFB1R1, TGFB2R1, TIGIT, TIPARP, TMEM127, TMPRSS2, TNFAIP3, TNFRSF14, TNFRSF17, TOP1, TOP2A, TP53BP1, TP63, TRAF3, TRAF7, TSHR, TYMS, TYRO3, U2AF1, UGT1A1, VSIR, WEE1, WNK1, WRN, WT1, XBP1, XPA, XPC, XPO1, XRCC1, XRCC2, YEATS4, ZFXH3, ZMYM3, ZNF217, ZNF703, ZNF750, ZNRF3, ZRSR2

Tempus Disclaimer

The analysis of nucleic acids by next-generation sequencing (NGS) can be affected by multiple factors including DNA quality, hemolysis of the peripheral blood sample, clonal hematopoiesis (CH), and low amounts of circulating cell-free DNA limiting sensitivity. Tempus may report findings below our sensitivity threshold due to reduced sample quality and/or quantity. For samples flagged as falling below this threshold, Tempus advises resequencing in order to provide more accurate results. Additionally, the chance of detecting genetic alterations may be reduced in regions of the genome which are structurally difficult to sequence, in homologous genes, or due to sequencing errors.

Genetic alterations are defined as clinically significant based on peer-reviewed published literature and other authoritative sources such as NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®). These references are not comprehensive, therefore clinically unknown findings may occur.

These test results and information contained within the report are current as of the report date. Tempus will not update reports or send notification regarding reclassification of genomic alterations. This test was developed and its performance characteristics determined by Tempus AI, Inc. It has not been cleared or approved by the US Food and Drug Administration. The laboratory is CLIA certified to perform high-complexity testing. Any decisions related to patient care and treatment choices should be based on the independent judgment of the treating physician and should take into account all information related to the patient, including without limitation, the patient and family history, direct physical examination and other tests. Tempus is not liable for medical judgment with regards to diagnosis, prognosis or treatment in connection with the test results.

Tempus Disclaimer (continued)

Note that certain tumor type-, sample- or variant-related characteristics, such as low cell-free DNA concentration, may result in reduced analytic sensitivity of the xF+ test for detection of alterations in the covered genes, including the above mentioned guideline-recommended genes.

Circulating tumor DNA analysis can assist in monitoring cancer patients as a prognostic biomarker in the literature, with lower levels of ctDNA tumor fraction correlated with better outcomes (1). Any clinical decision, including additional testing to monitor for cancer progression, should be made in the context of the patient's entire medical history, not on the results of ctDNA tumor fraction alone.

This assay cannot conclusively determine whether cell-free variants detected in plasma are derived from a patient's germline, from their solid tumor, or from clonal blood cells. Follow-up testing with whole blood is recommended to determine whether any detected clonal hematopoietic variants represent potential clonal hematopoiesis of indeterminate potential (CHIP), clonal hematopoiesis of oncogenic potential (CHOP), or hematologic neoplasms, if clinically indicated.

References

1. Reichert ZR, Morgan TM, Li G, et al. Prognostic value of plasma circulating tumor DNA fraction across four common cancer types: a real-world outcomes study. *PLoS One*. 2022;34(1):111-120. doi:https://doi.org/10.1016/j.annonc.2022.09.163
2. Bick AG, Weinstock JS, Nandakumar SK, et al. Inherited causes of clonal haematopoiesis in 97,691 whole genomes. *Nature*. 2020;586(7831):763-768. doi:https://doi.org/10.1038/s41586-020-2819-2
3. Bernstein N, Chapman MS, Nyamondo K, et al. Analysis of somatic mutations in whole blood from 200,618 individuals identifies pervasive positive selection and novel drivers of clonal hematopoiesis. *Nat Genet*. 2024;56(10):1147-1155. doi:https://doi.org/10.1038/s41588-024-01755-1
4. Sonnenschein A, Hasan M, Hui S, et al. Abstract 2324: Leveraging a comprehensive genomic data library for detecting clonal hematopoiesis in liquid biopsy. *Cancer Res*. 2024;84(6_Suppl):2324. doi:https://doi.org/10.1158/1538-7445.AM2024-2324

Dostarlimab is indicated for the treatment of patients with mismatch repair deficient (dMMR) endometrial cancer or solid tumors. Although patients with MSI-H cancers are likely dMMR, confirmatory testing is recommended.

Dates and times are represented in the coordinated universal time zone (UTC) unless otherwise specified. However, dates that are provided to Tempus without a timestamp (e.g., sample collection date) are listed as provided.

Correlation with tumor and/or blood sequencing results may be helpful in clarifying the origin of variants in some cases.

Referenced with permission from the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) Acute Lymphoblastic Leukemia, Version: 2.2024;; Acute Myeloid Leukemia, Version: 3.2024;; Ampullary Adenocarcinoma, Version: 2.2024;; Anal Carcinoma, Version: 1.2024;; Basal Cell Skin Cancer, Version: 3.2024;; B-Cell Lymphomas, Version: 3.2024;; Biliary Tract Cancers, Version: 4.2024;; Bladder Cancer, Version: 4.2024;; Bone Cancer, Version: 1.2025;; Breast Cancer, Version: 4.2024;; Castleman Disease, Version: 1.2024;; Central Nervous System Cancers, Version: 2.2024;; Cervical Cancer, Version: 4.2024;; Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma, Version: 3.2024;; Chronic Myeloid Leukemia, Version: 1.2025;; Colon Cancer, Version: 5.2024;; Dermatofibrosarcoma Protuberans, Version: 1.2024;; Esophageal and Esophagogastric Junction Cancers, Version: 4.2024;; Gastric Cancer, Version: 4.2024;; Gastrointestinal Stromal Tumors, Version: 2.2024;; Gestational Trophoblastic Neoplasia, Version: 2.2024;; Hairy Cell Leukemia, Version: 1.2025;; Head and Neck Cancers, Version: 4.2024;; Hepatobiliary Cancers, Version: 1.2023;; Hepatocellular Carcinoma, Version: 3.2024;; Histiocytic Neoplasms, Version: 2.2024;; Hodgkin Lymphoma, Version: 3.2024;; Kaposi Sarcoma, Version: 1.2024;; Kidney Cancer, Version: 2.2025;; Melanoma: Cutaneous, Version: 3.2024;; Melanoma: Uveal, Version: 1.2024;; Merkel Cell Carcinoma, Version: 1.2024;; Mesothelioma: Peritoneal, Version: 2.2024;; Mesothelioma: Pleural, Version: 1.2024;; Multiple Myeloma, Version: 1.2025;; Myelodysplastic Syndromes, Version: 3.2024;; Myeloid/Lymphoid Neoplasms with Eosinophilia and Tyrosine Kinase Gene Fusions, Version: 2.2024;; Myeloproliferative Neoplasms, Version: 2.2024;; Neuroblastoma, Version: 2.2024;; Neuroendocrine and Adrenal Tumors, Version: 2.2024;; Non-Small Cell Lung Cancer, Version: 10.2024;; Occult Primary, Version: 2.2025;; Ovarian Cancer/Fallopian Tube Cancer/Primary Peritoneal Cancer, Version: 3.2024;; Pancreatic Adenocarcinoma, Version: 3.2024;; Pediatric Acute Lymphoblastic Leukemia, Version: 1.2025;; Pediatric Aggressive Mature B-Cell Lymphomas, Version: 2.2024;; Pediatric Central Nervous System Cancers, Version: 1.2024;; Pediatric Hodgkin Lymphoma, Version: 1.2024;; Penile Cancer, Version: 1.2024;; Primary Cutaneous Lymphomas, Version: 3.2024;; Prostate Cancer, Version: 4.2024;; Rectal Cancer, Version: 4.2024;; Small Bowel Adenocarcinoma, Version: 5.2024;; Small Cell Lung Cancer, Version: 2.2025;; Soft Tissue Sarcoma, Version: 3.2024;; Squamous Cell Skin Cancer, Version: 1.2024;; Systemic Light Chain Amyloidosis, Version: 1.2025;; Systemic Mastocytosis, Version: 3.2024;; T-Cell Lymphomas, Version: 4.2024;; Testicular Cancer, Version: 1.2024;; Thymomas and Thymic Carcinomas, Version: 1.2024;; Thyroid Carcinoma, Version: 4.2024;; Uterine Neoplasms, Version: 3.2024;; Vaginal Cancer, Version: 2.2025;; Vulvar Cancer, Version: 4.2024;; Waldenström Macroglobulinemia/Lymphoplasmacytic Lymphoma, Version: 1.2025;; Wilms Tumor (Nephroblastoma), Version: 1.2024. All rights reserved. Accessed October 1, 2024. To view the most recent and complete version of the guideline, go online to [NCCN.org](https://www.nccn.org). The NCCN Guidelines® and other content provided by NCCN are works in progress that may be refined as often as new significant data becomes available. They are statements of consensus of its authors regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult any NCCN Guidelines® or other NCCN Content is expected to use independent medical judgment in the context of individual clinical circumstances to determine any patient's care or treatment. NCCN makes no warranties of any kind whatsoever regarding their content, use or application and disclaims any responsibility for their application or use in any way. Therapeutic options are not applicable in all disease settings.

The OncoKB™ precision oncology knowledge base was made available under license from Memorial Sloan Kettering Cancer Center. See terms [here](#).